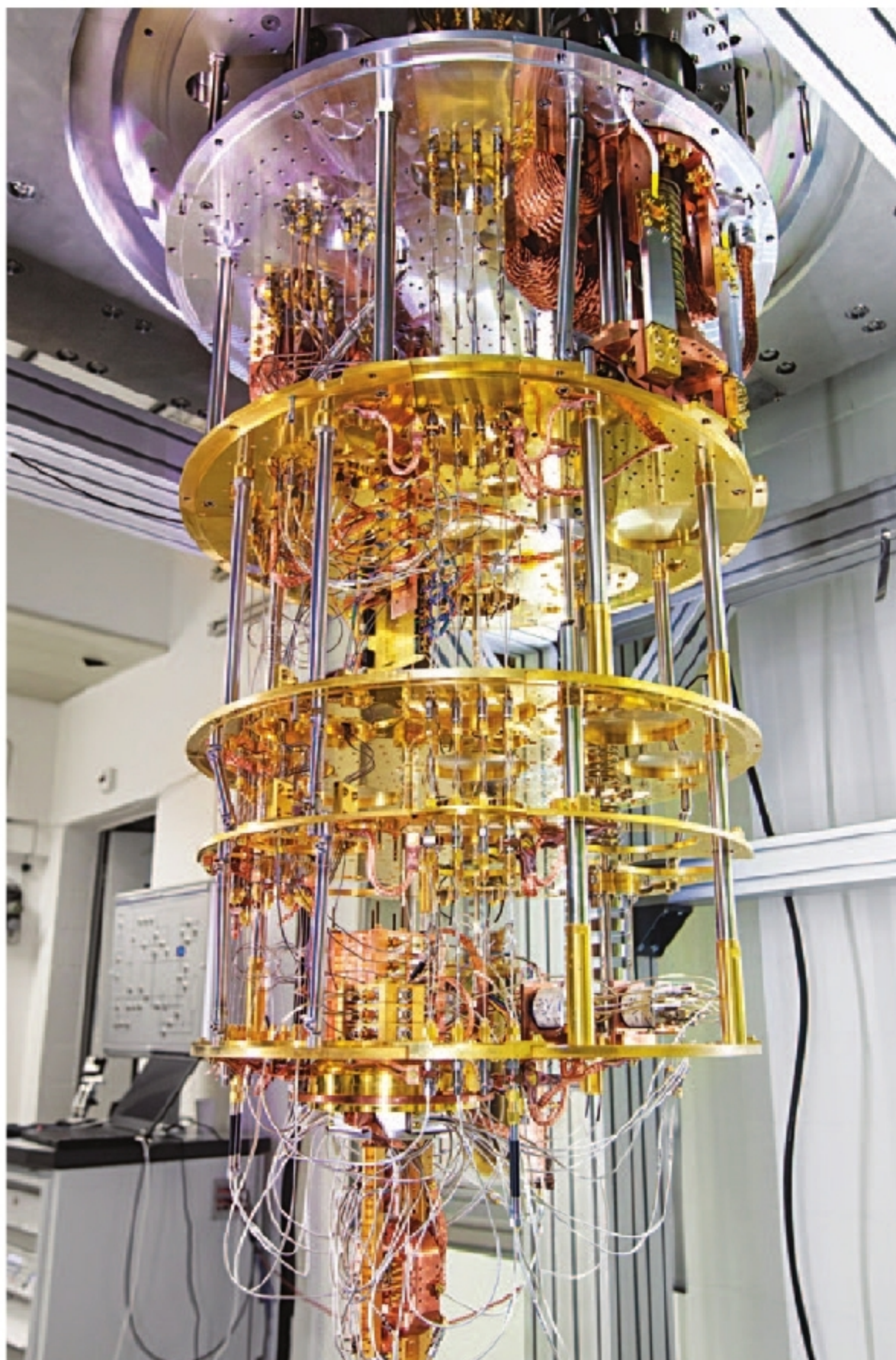


A Quick Look at Quantum Computing

Here's a brief introduction to quantum computing, covering the advances made in the recent past and the future aspects. This article also simplifies the world of quantum computing for the lay person and may ignite a spark of interest among readers that could lead to further reading in this domain.



Quantum computing is a comparatively new area of research and development. Towards the end of the 20th century, scientists decided to combine two of the most influential and revolutionary theories of the century — quantum mechanics and information theory. Their success led to a new perspective of computation and information. Information theory, which is an amalgamation of computer science as well as communications, swept through the world of physics, impacting it considerably. Hence, significant topics of information theory could be now discussed even without understanding the contingents of the physical devices used for communication or computation. This extremely powerful ability to ignore the underlying principles of physics can be seen in our day-to-day computation and communication devices.

Quantum mechanics has an important purpose in the evolution of the latest and most effective computing devices. Traditional computing devices operate on principles of quantum mechanics, from transistors to advanced hardware, that are responsible for speeding up computation while minimising device size. We can feel the charm of quantum mechanics in the low-level implementation realm. Until recently, it had absolutely nothing to do with the process of computation or communication, in any way.

In the early 1980s, a few researchers who believed that quantum mechanics might have implications in information theory in ways that were unanticipated